Aamir Mohammed Khan

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Profile

Fourth-year CSE undergraduate (CGPA 9.77/10) with hands-on AI-assisted development using Claude, Claude Code, and GitHub Copilot. Builds full-stack and LLM-powered applications; uses AI tools for implementation, debugging, testing, and code review while validating generated changes.

Education

M. S. Ramaiah University of Applied Sciences, Bengaluru 2023 – 2027
Bachelor of Technology in Computer Science and Engineering CGPA: **9.77/10.0**

Technical Skills

AI-Assisted Development: Claude, Claude Code, GitHub Copilot, ChatGPT, Gemini, Antigravity; implementation, codebase exploration, debugging, test generation, refactoring, code review

AI / APIs: LLM API integration, prompt engineering, hybrid retrieval (BM25 and embeddings), tool calling, structured outputs, server-sent events (SSE), JSON validation and repair

Languages: Python, C, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML, CSS

Frameworks and Libraries: React 19, Node.js, Express.js, FastAPI, Tailwind CSS, Streamlit, Pandas, scikit-learn, XGBoost

Data and Queues: MongoDB Atlas, PostgreSQL, Redis, BullMQ

Cloud, DevOps and Web: Docker, Linux, Azure VM, Render, Vercel, Git, GitHub Actions, CI/CD, REST APIs, Progressive Web Apps (PWA)

Core CS: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks

Experience

AlgoUniversity Remote | July 2026 – September 2026
Software Development Extern Entropy-OJ Live App

- Built a responsive coding workspace with **React 19, TypeScript, and Monaco Editor**, including mobile navigation, split-pane editing, compiler diagnostics, and submission history.
- Implemented asynchronous **C++17/Python 3.11** judging with Node.js/Express, **BullMQ/Redis**, and MongoDB Atlas; enforced Linux resource limits, execution timeouts, output caps, and per-job workspaces for untrusted submissions.
- Integrated three LLM workflows using **Gemini Flash with Groq/OpenRouter fallback**: progressive debugging hints, problem-setting QA, and post-acceptance complexity classification; added guardrails against code-spoiling hints and hidden-test leakage.
- Used Claude and Render logs to trace a **2-minute, 12-second submission delay** to Redis `ETIMEDOUT`; enabled TCP keep-alive, raised the Azure public-IP idle timeout, and verified prompt processing of subsequent submissions.

Technical Projects

AI Digital Assistant Portfolio – RAG-Powered Recruiter Chatbot Live App | Source Code

- Built a full-stack recruiter assistant with **FastAPI, React, TypeScript, and PostgreSQL**, using hybrid **retrieval-augmented generation (RAG)** with BM25 and embeddings over curated profile facts to ground responses.
- Implemented **server-sent events (SSE)** for streaming answers with source citations, human handoff for unsupported questions, and LLM tool calling with server-side argument validation for structured job-fit and skill-gap reports.
- Added prompt-injection defences, rate limiting, token-budget controls, model-provider failover, and a **GitHub Actions evaluation workflow** to assess retrieval and response quality.

HealthTracker – Health and Activity Tracking PWA Live App | Source Code

- Built a full-stack **Progressive Web App (PWA)** for nutrition, workout, and health-metric tracking with interactive dashboards; configured GitHub Actions CI/CD for build, test, and deployments to Vercel/Render, with MongoDB compound indexes and cache-first data access.

Health Premium Predictor – Machine Learning Application Live App | Source Code

- Developed and deployed a **Streamlit** insurance-premium predictor using **XGBoost regression**, feature engineering, missing-value imputation, and hyperparameter tuning; reported an **R^2 score of 0.99** and approximately **3% prediction error**.

Honours and Achievements

- **2nd Runner Up Team – Google Cloud Build with AI: Code for Communities, Second Edition:** Achieved 2nd Runners Up position in **Track 2: CleanAir & Clear Streets**. Hackathon details.
- **University-wide First Place – Graph Algorithm Contest (2025):** Earned first rank in a graph-algorithm contest led by a Codeforces Master and received a 50% scholarship to AlgoUniversity's Accelerator Program.